Day 5 — Git & GitHub

Topics

 Introduction to Git and GitHub

 Repository creation

 Git installation and configuration

 git init, git add, git commit

 git push, git pull, and git clone

 Branching basics

 README file

Tasks

 Create a GitHub account if required.

 Create a repository for internship tasks.

 Upload Days 1–4 programs.

 Write a README explaining the programs.

 Practice basic Git commands.

Expected Output

 GitHub repository created.

 Initial internship tasks uploaded.

1] Introduction to Git and GitHub

Git is a free, open-source Version Control System (VCS) used to track changes made to files in a project.

In simple words, Git remembers the history of your code. If you modify a file and something goes wrong, Git can help you review earlier versions and recover previous work.

2] Repository creation:

Repository creation is one of the first practical steps in learning Git and GitHub. Let's understand it step
git push, git pull, and git clone

1. What is a Branch in Git?

A branch in Git is a separate line of development that allows you to work on new features, fix bugs, or complete internship tasks without directly changing the main code.

Think of a branch as a separate workspace for your code.

Example: Suppose you have an internship repository named `internship-tasks`. You can create different branches for different tasks.

3]git push, git pull, and git clone

- `git push` — uploads your committed changes to GitHub.
- `git pull` — downloads new changes from GitHub and integrates them into your current branch.
- `git clone` — copies an existing remote repository to your computer.

Git branching is an important concept for Java Full Stack Developers because developers often work on different features of the same project without disturbing the main code.

Let's learn it step by step with simple English, Git commands, practical examples, and interview questions.

1. What is a Branch in Git?

A branch is an independent line of development in a Git repository. It lets you make changes to your code without directly changing the work on another branch.

Imagine you are developing a Student Management System. Your project has a working version, and now you want to add a login page.

1. What is a README file?

Definition: A README file is a document that provides information about a project, including its purpose, features, installation steps, usage, and requirements.

- `README` means “read me.”
- `.md` stands for Markdown, a lightweight formatting language.
- `README.md` is commonly displayed automatically on a repository's GitHub homepage.

Example: If you create a Java Student Management System, the README explains the project to another developer who wants to understand or run your code.

Practice basic Git commands

Command	Purpose
`git init`	Initialize a Git repository
`git status`	Check changed and untracked files
`git add .`	Stage changes
`git commit -m "message"`	Save a snapshot of staged changes
`git log --oneline`	View recent commits
`git branch`	List local branches
`git switch -c task-1`	Create and switch to a branch
`git push`	Upload commits to GitHub
`git pull`	Fetch and integrate remote changes
`git clone URL`	Copy a remote repository to your computer